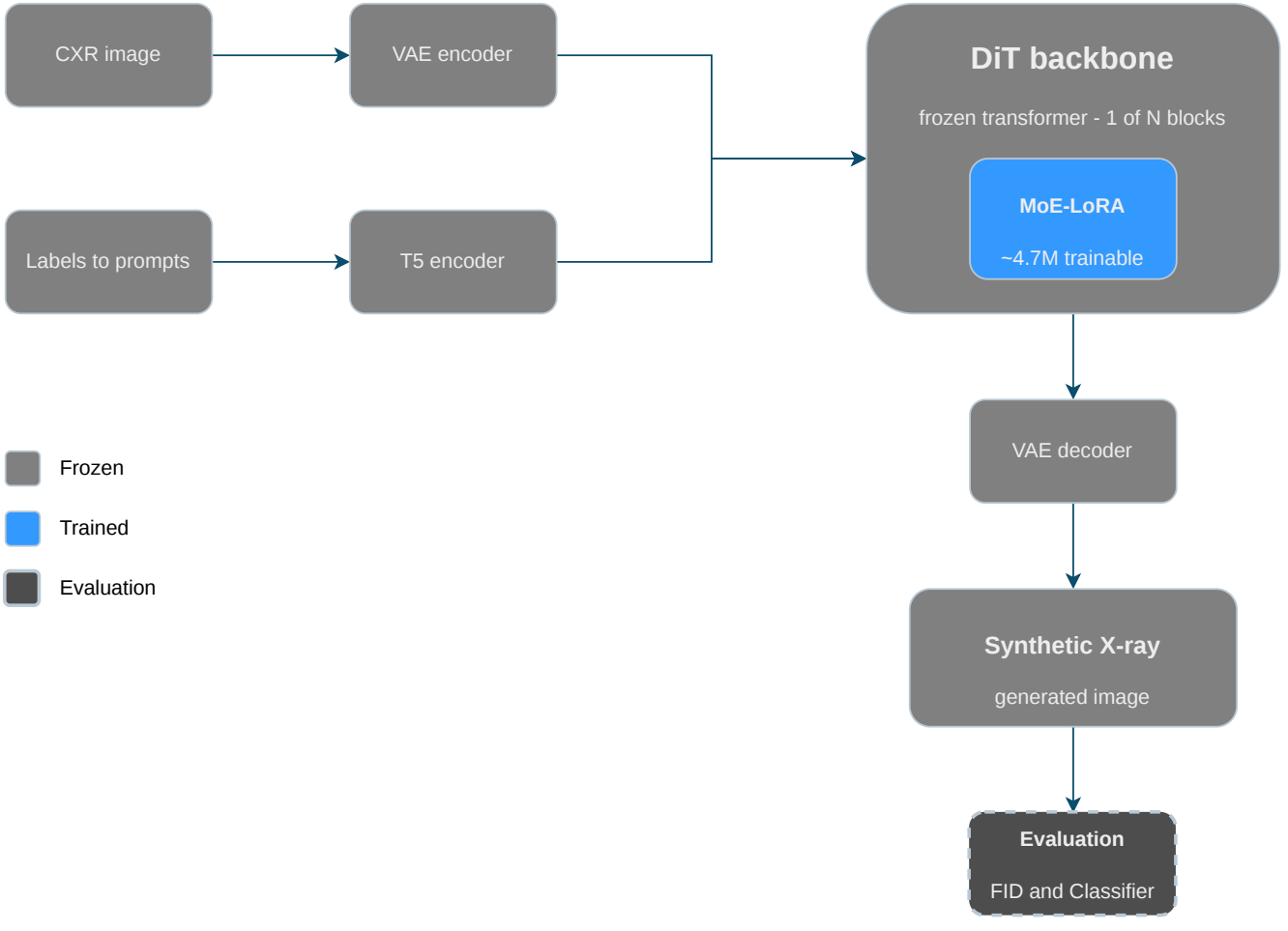
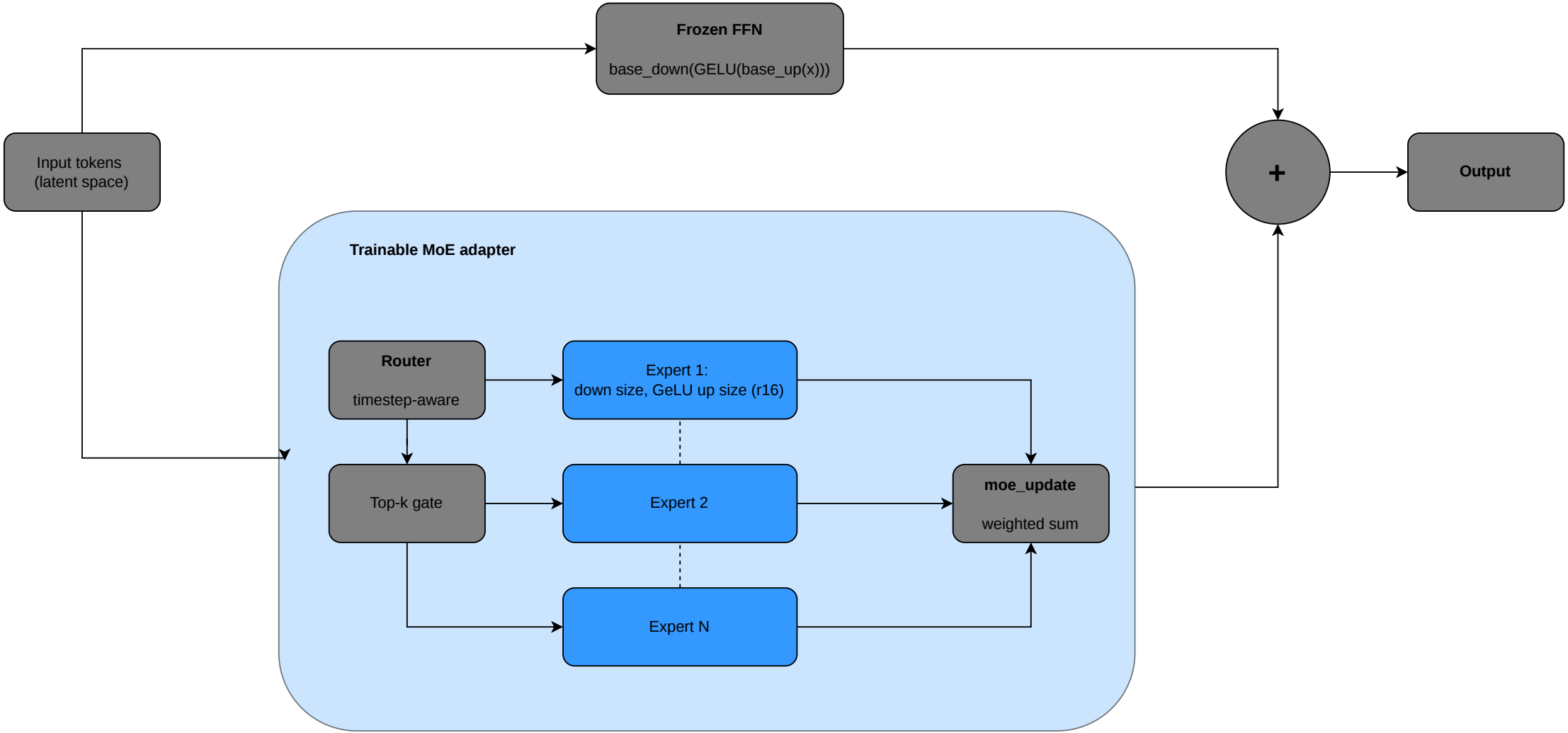
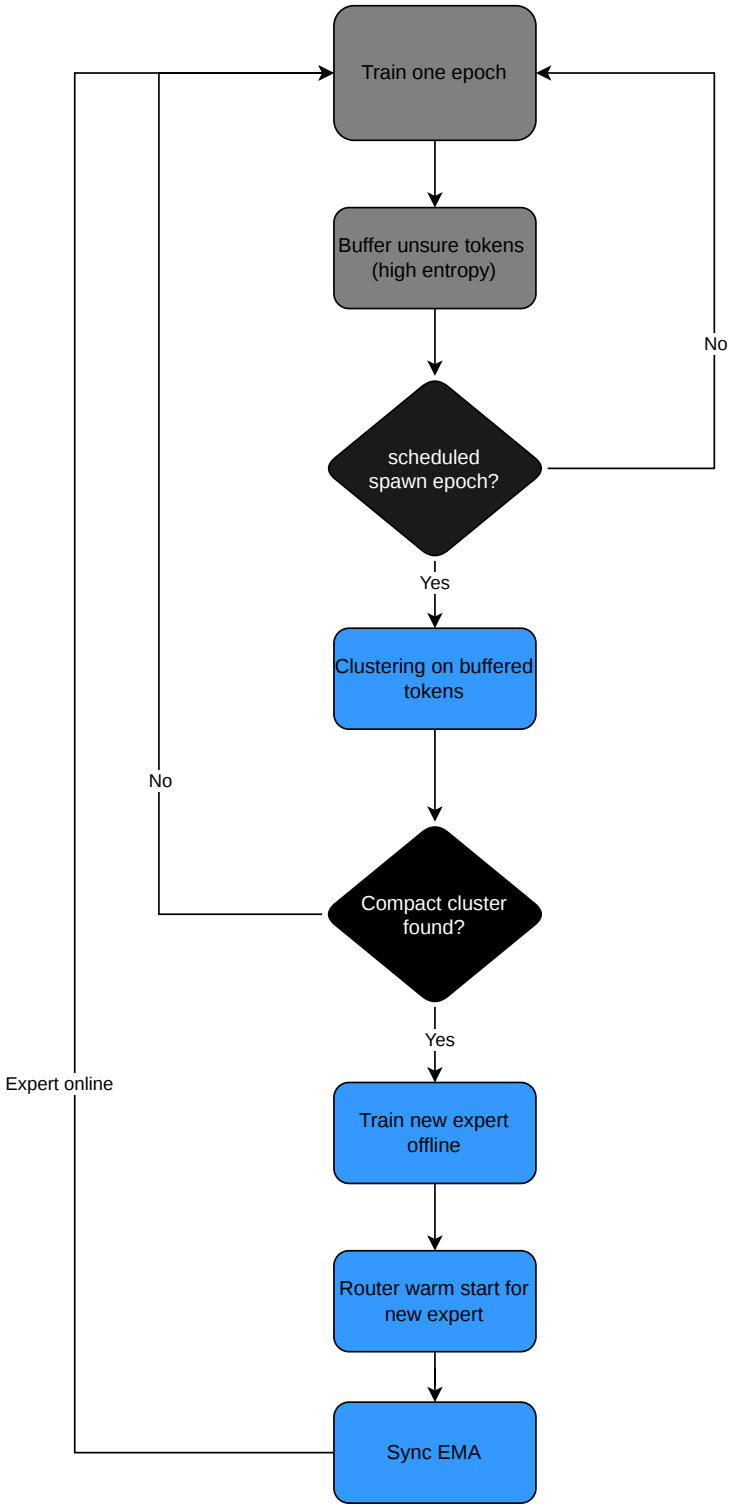

CXR image
VAE encoder
Labels to prompts
T5 encoder
DiT backbone
frozen transformer - 1 of N blocks
MoE-LoRA
~4.7M trainable
VAE decoder
Synthetic X-ray
generated image
Evaluation
FID and Classifier
Frozen
Trained
Evaluation


Frozen FFN
base_down(GELU(base_up(x)))
Input tokens
(latent space)
+
Output
Trainable MoE adapter
Router
timestep-aware
Expert 1:
down size, GeLU up size (r16)
Top-k gate
Expert 2
moe_update
weighted sum
Expert N


Train one epoch
Buffer unsure tokens (high entropy)
scheduled spawn epoch?
No
Yes
Clustering on buffered tokens
No
Compact cluster found?
Yes
Expert online
Train new expert offline
Router warm start for new expert
Sync EMA